

LIKENESS : THE GAME

V003 OPTIONAL EXPANSION MODULE / BLENDER MCP + FAL.AI ASSET FABRICATION

This document bakes the Blender MCP and fal.ai workflow into the existing OVS LIKENESS game pipeline as an optional fabrication layer. It strengthens asset creation, baking, PBR material generation, controlled shader tests, animation studies, and procedural environment dressing without replacing the core Unreal, Three.js, Image-Blaster, NanoGS, or OVS Production Bible foundation.

01 / Placement In The Pipeline

Blender is the fabrication lab. Unreal is the playable build. Three.js is the interactive companion. The OVS Production Bible remains the authority. This module lives between approved AI reference sheets and final engine import.

Use it after the 4-step reference system: Ideation, Clean Sheet, Parts, Views. Do not generate final game assets from loose concept art.

02 / Baseline Stack

Required: Blender connector access, Blender MCP connection, fal.ai MCP or approved generation provider, Git, staging folders, asset registry, and screenshot QA.

The supplied workflow identifies Blender connector plus fal.ai MCP as the two baseline pieces, with optional Skills added only when a specific use case requires them.

03 / Setup Gate

Confirm Blender MCP connection. Confirm fal.ai/provider connection when generation is needed. Confirm active scene is staging. Confirm Git status. Capture a before screenshot. Never commit API keys.

04 / Smoke Test

Start with a PBR material corner: floor, two walls, one table, four tileable materials, warm preview lighting, and one screenshot. This tests Blender control, provider access, material wiring, UV tiling, and screenshot verification without risking hero assets.

05 / Approved Workflow Patterns

Bake high-poly to low-poly. Assemble draft character GLBs from approved images. Build PBR material test corners. Test toon/outline shader passes. Test iridescent materials. Animate multi-stage prop assemblies. Use Geometry Nodes for procedural dressing.

06 / LIKENESS Asset Targets

Room 14 wall/floor/ceiling materials, bathroom tile, mirror distortion, brass key 14, camcorder, CRT/VHS stack, desk bell, chain lock, black device, locket, photo frames, parking-lot gravel, wet asphalt, Dellwood signage, dust/debris/photo scatter systems.

07 / Promotion Rules

Generated or repaired assets stay in staging until reviewed. Promotion requires clean mesh, correct scale, clean UVs, named materials, texture maps, collision proxy when needed, clean import test, screenshot proof, and registry update.

08 / Web And Unreal Split

For Unreal, prefer FBX for skeletal or collision-heavy assets and GLB for quick prototype props. Use _COL suffix for collision meshes. For Three.js, prefer optimized GLB with low material count and lightweight texture budgets.

09 / Copy-Paste Agent Prompt

You are the OVS asset fabrication agent for LIKENESS : THE GAME. Read README_START_HERE.md, docs/03_Character_Reference_4Step_Template.md, docs/07_Art_Direction_Style_Bible.md, docs/08_QA_QC_Checklists.md, docs/16_Blender_MCP_FAL_Asset_Fabrication_Module.md, data/asset_manifest.csv, and data/blender_asset_workflow_registry.csv. Confirm Blender MCP connection, provider connection if needed, staging scene status, Git status, and before screenshot. Do not reset scenes, overwrite approved assets, expose API keys, invent character identity, or export directly to final engine folders. Build, repair, bake, materialize, animate, scatter, export to staging, screenshot, report created files, and suggest registry updates.

10 / Prompt Pack Index

ID	Prompt File	Use
01	01_bake_high_to_low.md	Normal/AO bake from high-poly to low-poly
02	02_generate_draft_character_from_images.md	Draft GLB assembly from approved refs
03	03_pbr_material_test_corner.md	PBR material smoke test and lookdev
04	04_controlled_toon_outline_test.md	Controlled stylization experiment
05	05_procedural_iridescent_test.md	Iridescent test material
06	06_multistage_prop_animation.md	Key/tape/prop assembly animation
07	07_geometry_nodes_scatter_dressing.md	Dust/debris/photo/gravel scatter systems

11 / Human Review Gate

Ask: does it still feel like LIKENESS, does the object serve a story or mechanics purpose, does it improve the game rather than just show off tooling, is it technically clean, and can Unreal and Three.js share it without style drift?

12 / V003 Package Additions

- docs/16_Blender_MCP_FAL_Asset_Fabrication_Module.md
- agents/expansion_prompts/BLENDER_MCP_FAL_AGENT_PROMPT.md
- data/blender_asset_workflow_registry.csv
- expansions/blender_mcp_fal/README_BLENDER_MCP_FAL.md
- expansions/blender_mcp_fal/prompts/*.md
- expansions/blender_mcp_fal/tests/BLENDER_ASSET_QA_CHECKLIST.md